

11 Residual networks

11.1

$$\begin{aligned} y &= h_3 + f_4[h_3, \phi_4] \\ &= h_2 + f_3[h_2, \phi_3] + f_4[h_3, \phi_4] \\ &= h_1 + f_2[h_1, \phi_2] + f_3[h_2, \phi_3] + f_4[h_3, \phi_4] \\ &= x + f_1[x, \phi_1] + f_2[h_1, \phi_2] + f_3[h_2, \phi_3] + f_4[h_3, \phi_4] \\ &= x + f_1[x, \phi_1] \\ &\quad + f_2[x + f_1[x, \phi_1], \phi_2] \\ &\quad + f_3[x + f_1[x, \phi_1] + f_2[x + f_1[x, \phi_1], \phi_2], \phi_3] \\ &\quad + f_4[x + f_1[x, \phi_1] + f_2[x + f_1[x, \phi_1], \phi_2] + f_3[x + f_1[x, \phi_1] + f_2[x + f_1[x, \phi_1], \phi_2], \phi_3], \phi_4] \end{aligned}$$

11.2 simple combinatorics: U nodes, length l : $\binom{U}{l}$ paths,

e.g. $\binom{3}{0} = 1, \binom{3}{1} = 3, \binom{3}{2} = 3, \binom{3}{3} = 1$

$$\binom{5}{0} = 1, \binom{5}{1} = 5, \binom{5}{2} = 10, \binom{5}{3} = 10, \binom{5}{4} = 5, \binom{5}{5} = 1$$

11.3 let $s_1 = f_1[x], s_2 = f_2[x + s_1], s_3 = f_3[x + s_1 + s_2]$

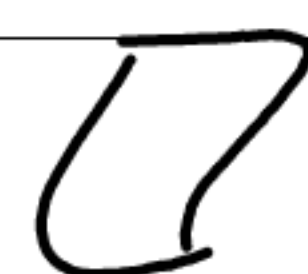
$$s_4 = f_4[x + s_1 + s_2 + s_3], y = x + s_1 + s_2 + s_3 + s_4$$

$$\frac{\partial y}{\partial h_1} = \frac{\partial y}{\partial s_1} = \frac{\partial x}{\partial s_1} + \frac{\partial s_1}{\partial s_1} + \frac{\partial s_2}{\partial s_1} + \frac{\partial s_3}{\partial s_1} + \frac{\partial s_4}{\partial s_1}$$

$$= I + \frac{\partial f_2}{\partial h_1} + \frac{\partial s_3}{\partial s_1} + \frac{\partial s_4}{\partial s_1}$$

By the chain rule, $\frac{\partial s_3}{\partial s_1} = \frac{\partial f_3}{\partial h_1} + \frac{\partial s_3}{\partial s_2} \frac{\partial s_2}{\partial s_1} = \frac{\partial f_3}{\partial h_1} + \frac{\partial f_3}{\partial h_2} \frac{\partial h_2}{\partial h_1}$

and $\frac{\partial s_4}{\partial s_1} = \frac{\partial f_4}{\partial h_1} + \frac{\partial f_4}{\partial h_3} \frac{\partial h_3}{\partial h_2} \frac{\partial h_2}{\partial h_1} + \frac{\partial f_4}{\partial h_3} \frac{\partial h_3}{\partial h_1} + \frac{\partial f_4}{\partial h_2} \frac{\partial h_2}{\partial h_1}$



11.4 As $\text{Var}(\bar{x+y}) = \text{Var}[X] + \text{Var}[Y]$ if A, B uncorrelated:

$$\begin{aligned}\text{Var}(\bar{x+y}) &= E[(x+y - E(\bar{x+y}))^2] \\&= E[(x+y - E[x] - E[y])^2] \\&= E[(x - E[x] + y - E[y])^2] \\&= E[(x - E[x])^2 + 2(x - E[x])(y - E[y]) \\&\quad + (y - E[y])^2] \\&= E[(x - E[x])^2] + 2E[(x - E[x])(y - E[y])] \\&\quad + E[(y - E[y])^2] \\&= \text{Var}(x) + \text{Var}(y) + \underbrace{2\text{Cov}[x, y]}_{0 \text{ if uncorrelated!}}\end{aligned}$$

Since the weight matrices are i.i.d. $\sim N(0, \cdot)$, and from independence follows uncorrelatedness, the values in the two branches are uncorrelated.

11.6 FCN without BatchNorm: $|\text{biases}| = 200 + 1$

$$|\text{weights}| = 20 + 3600 + 20 = 3640$$

$$|\text{total}| = 3841$$

FCN with BatchNorm: $|\text{offsets}| = 10 \cdot 20 = 200$

$$|\text{scale}| = 10 \cdot 20 = 200$$

$$+ |\text{total}| = 4241$$

When using BatchNorm, we can omit the hidden biases, as BatchNorm cancels out any constant shifts (biases) anyway! Thus, we end up with 4041 parameters.

11.7 If we only penalize the convolutional weights, BatchNorm's scaling parameters should tend to grow and effectively compensate, diluting the effect of ℓ_2 regularization!

11.8 Ignoring biases (BatchNorm makes them redundant):

$$\begin{aligned} \text{1st block: } |C_{in}| \times |C_{out}| \times |kernel| &= 512 \cdot 512 \cdot 3 \cdot 3 \\ &+ |BatchNorm| = 2 \cdot 512 \end{aligned}$$

$$\begin{aligned} &|convolution| + |BatchNorm| \\ \text{2nd block: } 1 \times 1 \text{ conv: } &512 \cdot 128 + 2 \cdot 512 \\ &3 \times 3 \text{ conv: } 128 \cdot 128 \cdot 3 \cdot 3 + 2 \cdot 128 \\ &1 \times 1 \text{ conv: } 128 \cdot 128 + 2 \cdot 128 \end{aligned}$$

11.9 Because BatchNorm statistic estimates become unstable!